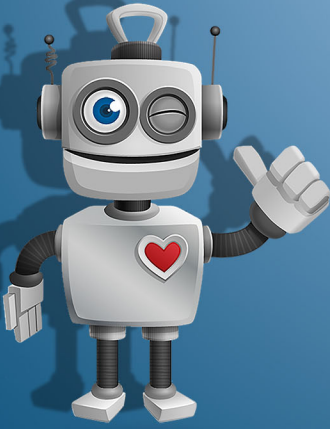


تقدم لكم اللجنة الاكاديمية في فريق

Mechatronics spectrum

اسئلة سنوات لمادة

أجهزة الحماية



ميد



www.spectrum-fet.weebly.com



[groups/mechatronics.groups](https://www.facebook.com/groups/mechatronics.groups)



0785060940

Q1: Circle the right answer then fill the tale below (17points)

1. Which among these are the main characteristics of a fuse element?
 - a. Low melting point
 - b. High conductivity
 - c. Least deterioration due to oxidation
 - d. All of the
2. Which among the following statement is true related to fuse?
 - a. Greater the current smaller is the time taken by the fuse to blow out.
 - b. Greater the current greater is the time taken by the fuse to blow out.
 - c. The current is directly proportion to the blow out time of fuse.
 - d. Is dependent on the temperature and atmospheric conditions.
3. For a current upto 10A which material is used as the fusing element
 - a. Copper
 - b. Silver
 - c. Alloy of lead and tin
 - d. Zinc
4. What is fusing factor?
 - a. The ratio of current rating of the fuse to the minimum fusing current.
 - b. The ratio of minimum fusing current to the current rating of the fuse.
 - c. The ratio of maximum fusing current to the current rating of the fuse.
 - d. The ratio of minimum fusing current to the voltage rating of the fuse
5. What should be the value of fusing factor?
 - a. Equal to zero
 - b. Equal to one
 - c. Less than one
 - d. More than one
6. A circuit breaker is
 - a- power factor correcting device
 - b- device to neutralize the effect of transients
 - c- waveform correcting device
 - d current interrupting device.
7. The function of protective relay in a circuit breaker is
 - a-to each any stray voltages
 - b-to close the contacts when the actuating quantity reaches a certain predetermined value
 - c-to limit arcing current during the operation of circuit breaker
 - d-to provide additional safety in the operation of circuit breaker.
8. The fault clearing time of a circuit breaker is usually
 - a-few minutes
 - b-few seconds
 - c-one second
 - d-few cycles of supply voltage.
9. Which of the following is not a type of the contactor for circuit breakers ?
 - (A) Electro-magnetic
 - (B) Electro-pneumatic
 - (C) Pneumatic
 - (D) Vacuum.
10. Miniature Circuit Breaker is a small
 - a-fuse
 - b-magnetic switch
 - c-electromagnetic switch
 - d-two way switch
11. Using a high current fuse in a low current appliance is very
 - a-safe
 - b-dangerous
 - c-necessary
 - d-complicated

14

4

3.5

7

29

12. Circuit breakers work
a-constantly
c-when there is resistance

b-only once
d-when there is magnetic field

13. A simple enclosed length of wire that has a low melting point is known as a
a-resistor b-fuse
c-circuit breakers d-thermistor

14. A thermal protection switch provides protection against what?
a. Overload b. Temperature.
c. Short circuit. d. Over voltage

15. A relay used for protection of motors against overload is
(A) Impedance relay (B) Electromagnetic attraction type
(C) Thermal relay (D) Buchholz's relay.

16. A fuse wire should have
a-Low specific resistance and high melting point b-Low specific resistance and low melting point
c-High specific resistance and high melting point d-High specific resistance and low melting point.

17. The number of cycles in which a high speed circuit breaker can complete its operation is
(A) 3 to 8 (B) 10 to 18
(C) 20 to 30 (D) 40 to 50.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
d	a	b	b	c	d	b	d	d	b	b	d	b	a	b	b	a

Q2: What is the meaning of the commonly used abbreviations (5 points)

ALM	Alarm ✓
NO	Normally Open ✓
SEC	Secondary ✓
MSP	Motor Starter protection ✓
SEC	Secondary ✓
DPDT	Double pole Double Throw ✓
CAP	Capacitor ✓
TD	Time delay ✓
SP	Single pole ✓
OL	Over load ✓
SS	Selector switch ✓

Q3: List the three types of switches for industrial control and their branch types; give an example
(8 points)

A) magnetic switch:-

1) Knife

2) toggle

3) push button

B) electromagnetic switch:-

1) Solenoid

2) Relay

3) Semi conductor

3) mechanical switch:-

1) fuse

2) Snap acting

3) Limit Switch

4) mercury

Q1:

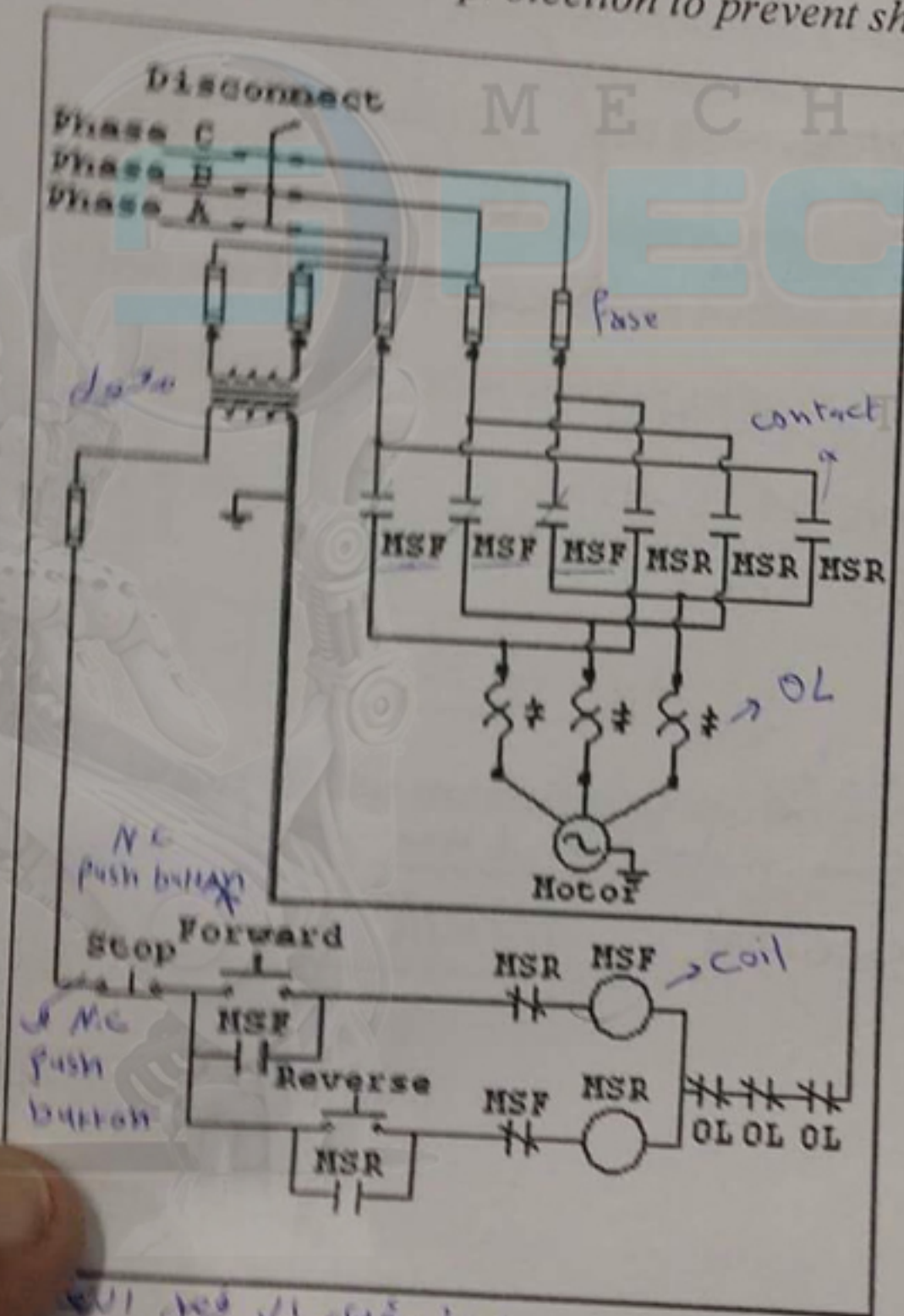
(10points)

Explain how this diagram work

The diagram's components

For what purpose this circuit is

The main electrical protection to prevent short circuit are.



المفتاح (push button) يصبح
[MSF] في حالة تفضية مما يؤدي

من [MSF] من N.O الى N.C

الى تشغيل المحرك

تشغيل المحرك في حالة Reverse

ضغط على المفتاح [Reverse]

توصيل الى coil [MSR] بالتيار

تحويل الى contact [MSR] من N.O

يؤدي الى عكس التدرج من الفان A

الى المحرك مما يؤدي الى تحويله

Rev

جدة إشارة المحرك فان علامات [OL]

سوف تصبح في حالة N.O

دعنا نزيد اطفاء المحرك نضغط على المفتاح stop الذي يعمل القاطع

transformer - fuse - contact - push button - coil - over load

هو توصيل المحرك في حالة ال Forward في حالة ال Reverse